

cp_4b

1.

Print the Score | 🏆 20/20

Easy

Write a program that reads a distance D in km and calculates the score.

- If D is less than or equal to **10**, the score is D .
- If D is greater than **10**, the score is the sum of **10** and $(D - 10) * 3$.

Input

The input will be a single line containing an integer representing D .

Output

The output should be a single line containing an integer that is the score.

Explanation

For example, if the given distance is $D = 3$,

- **3** is less than or equal to **10**
- The score is **3** (D)

The output should be **3**.

For example, if the given distance is $D = 25$,

- **25** is greater than **10**.
- The score is the sum of **10** and $(D - 10) * 3$.

```
10 + (D - 10) * 3
10 + (25 - 10) * 3
10 + (15) * 3
10 + 45
55
```

The output should be **55**

Sample Input 1

3

Sample Output 1

3

code:

```
d=int(input())
if(d <= 10):
    print(d)
else:
    d=10+(d-10)*3
    print(d)
```

2.

Cube of a Number | 🟡 15/15

Easy

Write a program that reads a number N and prints the **Cube** of N (N^3).

Input

The input will be a single line containing an integer representing N .

Output

The output should be a single line containing an integer that is the **Cube** of N (N^3).

Explanation

For example, if the given number is $N = 4$,

- The **Cube** of N is **64** ($4^3 = 64$).

The output should be **64**.

Sample Input 1

4

Sample Output 1

64

Sample Input 2

10

Sample Output 2

1000

code:

```
n=int(input())
print(n**3)
```

3.

Exponent of a Number | 🏆 20/20

Write a program that reads a number `N` and an exponent `E` and prints the result of `N` power `E` (N^E).

Input

The first line of input contains an integer representing the number `N` .
The second line of input contains an integer representing the exponent `E` .

Output

The output should be a single line containing an integer that is the result of `N` power `E` (N^E).

Explanation

For example, if the given number is `N = 3` and the exponent is `E = 2` ,

- The result of `N` power `E` is **9**. ($3^2 = 9$)

The output should be **9**.

For example, if the given number is `N = 5` and the exponent is `E = 3` ,

- The result of `N` power `E` is **125**. ($5^3 = 125$)

The output should be **125**.

Sample Input 1

```
3
2
```

Sample Output 1

```
9
```

Sample Input 2

```
5
3
```

Sample Output 2

```
125
```

code:

```
num=int(input())
exponent=int(input())
print(num**exponent)
```

4.

Square Root of a Number | 🏆 25/25

Easy

Write a program that reads a number `N` and prints the **Square Root** of `N`.

Note

To calculate the **Square Root** of `N`, use the `N` power **0.5** ($N^{0.5}$).

Input

The input will be a single line containing an integer representing `N`.

Output

The output should be a single line containing a float that is the **Square Root** of `N`.

Explanation

For example, if the given number is `N = 4`,

- The **Square Root** of `N` is **2.0** ($4^{0.5} = 2.0$).

The output should be **2.0**.

Sample Input 1

4

Sample Output 1

2.0

Sample Input 2

100

Sample Output 2

10.0

code:

```
n=int(input())
print(n**0.5)
```

5.

Sum of Squares | 🟡 25/25

● Solvec

Easy

Write a program that reads two numbers `A` and `B`, and checks if $A^2 + B^2$ (sum of the square of `A` and the square of `B`) is greater than or equal to 60.

Print **Greater than or Equal to 60** if the sum of the square of `A` and the square of `B` is greater than or equal to 60. Otherwise, print **Less than 60**.

Input

The first line of input contains an integer representing `A`.

The second line of input contains an integer representing `B`.

Output

The output should be a single line containing a string. **Greater than or Equal to 60** should be printed if $A^2 + B^2$ is greater than or equal to 60. Otherwise, **Less than 60** should be printed.

Explanation

For example, if the given numbers are `A = 10` and `B = 2`,

- The square of `A` is 100 (10^2).
- The square of `B` is 4 (2^2).
- The sum of the square of `A` and the square of `B` is 104 ($100 + 4 = 104$).
- 104 (sum of the square of `A` and the square of `B`) is greater than or equal to 60.

The output should be **Greater than or Equal to 60**.

Sample Input 1

```
10
2
```

Sample Output 1

```
Greater than or Equal to 60
```

Sample Input 2

```
1
3
```

Sample Output 2

```
Less than 60
```

code:

```
a=int(input())
b=int(input())
cond1=a**2 +b**2
if(cond1< 60):
    print("Less than 60")
else:
    print("Greater than or Equal to 60")
```

6.

Greatest Among The Exponents | 🟡 25/25

Write a program that reads two numbers A and B and finds the,

- Result of A power B (A^B)
- Result of B power A (B^A)

Print the greatest among the results of A power B (A^B) and B power A (B^A).

Input

The first line of input contains an integer representing A .

The second line of input contains an integer representing B .

Output

The output should be a single line containing an integer. The result of A^B should be printed if A^B is greater than B^A . Otherwise, the result of B^A should be printed.

Explanation

For example, if the given two numbers are $A = 2$ and $B = 3$,

- Result of A power B (A^B) is 8. ($2^3 = 8$)
- Result of B power A (B^A) is 9. ($3^2 = 9$)
- Greatest among the two results is 9. (9 is greater than 8)

The output should be 9.

Sample Input 1

```
2
3
```

Sample Output 1

```
9
```

Sample Input 2

```
3
1
```

Sample Output 2

```
3
```

code:

```
a=int(input())
b=int(input())
a_power_b=a**b
b_power_a=b**a
if(a_power_b>b_power_a):
    print(a_power_b)
else:
    print(b_power_a)
```

7.

Last Digit of a Square | 20/20

● Solved

Easy

Write a program that reads a number `N` and checks if the last digit of `N` is equal to the last digit of the square of `N`.
Print **Equal** if the last digit of `N` is equal to the last digit of the square of `N`. Otherwise, print **Not Equal**.

Input

The input will be a single line containing an integer representing `N`.

Output

The output should be a single line containing a string. **Equal** should be printed if the last digit of `N` is equal to the last digit of the square of `N`. Otherwise, **Not Equal** should be printed.

Explanation

For example, if the given number is `N = 15`,

- The square of `N` is **225** ($15^2 = 225$).
- The last digit of `N` (**15**) is **5**.
- The last digit of square of `N` (**225**) is **5**.
- **5** (last digit of **15**) is equal to **5** (last digit of **225**).

The output should be **Equal** as the last digit of `N` is equal to the last digit of the square of `N`.

Sample Input 1

15

Sample Output 1

Equal

Sample Input 2

2

Sample Output 2

Not Equal

code:

```
n=input()
num=int(n)**2
num_str=str(num)
if(n[-1]==num_str[-1]):
    print("Equal")
else:
    print("Not Equal")
```

8.

Armstrong Number | 20/20

● Sol

Medium

Write a program that reads a three-digit number `N` and checks if `N` is an **Armstrong Number**.

Note

Armstrong Number

A number is said to be an **Armstrong Number** if the sum of powers of all the digits of the number is the number itself.

Example:

371 is an **Armstrong Number** as the sum of the powers of all digits of `N` is equal to 371. ($3^3+7^3+1^3 = 371$)

Input

The input will be a single line containing a three-digit integer representing `N`.

Output

The output should be a single line containing a boolean. **True** should be printed if `N` is an **Armstrong Number**. Otherwise, **False** should be printed.

Explanation

For example, if the given number is `N = 371`,

- The digits in 371 are 3, 7 and 1.
- Number of digits in 371 is 3.
- Sum of digits to the power of 3 is 371. ($3^3+7^3+1^3 = 371$)
- 371 (Sum of the powers of all digits of `N`) is equal to 371 (`N`). So, `N` is an **Armstrong Number**.

The output should be **True**.

Sample Input 1

371

Sample Output 1

True

Sample Input 2

351

Sample Output 2

False

code:

```
n=input()  
n1=int(n[0])**3  
n2=int(n[1])**3  
n3=int(n[2])**3  
sum=n1+n2+n3  
sum_str=str(sum)  
if(sum_str==n):  
    print("True")
```

```
else:
    print("False")
```

9.

Divisible by Number - 2 | 🏆 25/25

Easy

Write a program that reads a number N and checks if the triple of N is divisible by 6.
Print the triple of N if the triple of N is divisible by 6. Otherwise, print N .

Note

The Triple of N is calculated as $3 * N$.

Example: Triple of 6 is 18. ($3 * 6 = 18$)

Input

The input will be a single line containing an integer representing N .

Output

The output should be a single line containing an integer. Triple of N should be printed if the triple of N is divisible by 6. Otherwise, N should be printed.

Explanation

For example, if the given number is $N = 6$,

- The triple of N is 18. ($3 * 6 = 18$)
- 18 is divisible by 6.

The output should be 18 ($3 * 6 = 18$).

For example, if the given number is $N = 9$,

- The triple of N is 27. ($3 * 9 = 27$)
- 27 is **not** divisible by 6.

The output should be 9 (N).

Sample Input 1

6

Sample Output 1

18

Sample Input 2

9

Sample Output 2

9

code:

```
n=int(input())
n_triple=n*3
if(n_triple%6==0):
    print(n_triple)
else:
    print(n)
```